Technical Documentation: AEGIS Orbital Debris Mitigation System

Project Name: AEGIS (Santa Industries)

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1. Project Overview

The AEGIS system is an automated solution designed where the density of objects in Low Earth Orbit (LEO) is high enough that collisions between objects could cause a cascade, making space activity and the use of satellites difficult or impossible for generations.

The system utilizes a combination of AI-driven radar tracking and laser-based systems to protect critical orbital infrastructure,

2. Technical Architecture

High-precision lasers are deployed to either vaporize small fragments or "nudge" larger pieces to degrade their orbit, causing them to safely burn up in the Earth's atmosphere.

3. Hardware Specifications (MVP)

The current Prototype (Minimum Viable Product) is built using the following components:

Controller: Micro:bit (Process Logic & User Interface).
Breakout Board: ELECFREAKS Wukong (Power management).
Actuators: Dual Servo Motors (High-torque servos for 2-axis movement).
Emitter: 5mW Laser Pointer (Simulating the high-powered neutralization laser).
Power: Integrated Li-ion battery via the Wukong board.

4. Electrical Wiring & Pin Mapping

To ensure consistent operation, the following pin configuration is utilized on the Wukong board:

Servo 1 (Horizontal mode): Connected to S0. Controls horizontal rotation.

Servo 2 (Vertical mode): Connected to S1. Controls vertical altitude.

Laser Emitter: Connected to P0 (GVS pins). Controlled via digital signal for firing sequences.

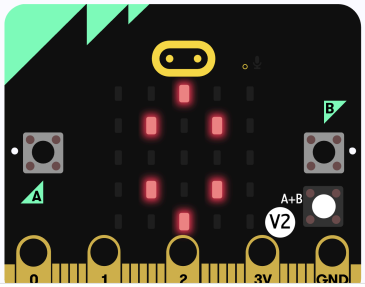
5. Software Logic & Controls

The software is written for the Micro:bit environment to manage the targeting system.

5.1 System Modes

The system features a Dual-Mode Toggle to allow precision targeting across two axes:

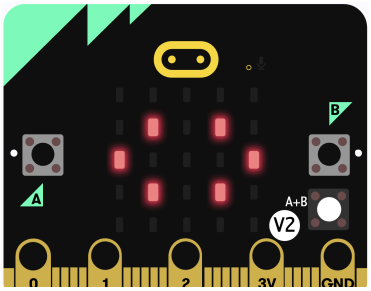
Horizontal Mode: Adjusts the horizontal angle of the laser assembly. LED with arrow pointing Up and down



Vertical Mode: Adjusts the vertical elevation of the laser assembly. LED with arrow pointing Left and right

5.2 Control Interface

Button A: Decreases the angle (Moves Left or Down depending on mode).



Button B: Increases the angle (Moves Right or Up depending on mode).

Buttons A+B: Toggles the active mode between Horizontal and Vertical mode.

5.3 Safety & Reliability Features

Mode Feedback: The Micro:bit LED grid provides a visual indicator of the current active axis to prevent operator error during manual calibration.

Software Limiter: The code includes predefined degree limits (e.g., 0° to 180°) to prevent the servo motors from straining against the physical chassis or tangling the laser wiring.

Sound indicator: The sound indicates the movement and mode switch, providing audio indictor of each movement.

6. Operation Guide

Initialization: Insert the Micro:bit into the Wukong board and power on.

Calibration: Use the A+B toggle to select the Tilt axis. Align the laser with the simulated orbital plane.

Targeting: Toggle to Pan mode and use Buttons A or B to track the simulated debris interception point.

Neutralization: Once confirmed alignment, the laser simulates the vaporization or orbital degradation process.

7. Safety Warning

Laser Safety: The laser emitter used in this MVP is for simulation purposes only. Do not point the beam at eyes, reflective surfaces, or aircraft.

Power Management: Ensure the Wukong board is switched off when not in use to preserve battery health.